Does Crude Oil Price Volatility Affect Inflation in India?

An Econometric Analysis of Indian CPI Inflation, 2014–2025

Econometric Analysis Laboratory Project

Abstract

This study examines whether crude oil price changes and volatility are associated with Indian CPI inflation. Monthly observations cover January 2014–December 2025; the final dynamic regression uses 132 observations from January 2015–December 2025. Contemporaneous oil-price changes and contemporaneous three-month rolling oil volatility are statistically insignificant in the tested models. One-month-lagged oil volatility is positive (0.015053) and statistically significant with Newey–West HAC standard errors (Bartlett kernel, bandwidth 4; p = 0.0046). Significant residual autocorrelation remains, so the result is a conditional association rather than evidence of causality.

1. Introduction and Research Objective

Crude oil is an important input into transport, production, and distribution. International oil-market movements may affect domestic prices through fuel costs, transport costs, exchange rates, taxes, and pricing policies. Price direction and volatility are distinct: direction measures whether prices rise or fall, while volatility measures recent fluctuations.

This study examines whether crude oil price volatility is associated with Indian CPI inflation and whether any association is contemporaneous or appears with a one-month delay. It is exploratory and does not identify a causal effect.

2. Data and Variables

The analysis-ready sample spans January 2014–December 2025 (144 observations). The dependent variable is combined Indian CPI year-on-year inflation. The final dynamic model uses 132 observations from January 2015–December 2025 due to lagged regressors.

Explanatory variables include monthly percentage change in crude oil prices, three-month rolling standard deviation of monthly oil log returns, and monthly percentage change in the USD/INR exchange rate. The volatility measure is a backward-looking proxy, not a GARCH-estimated conditional variance.

3. Methodology

The analysis uses descriptive statistics, time-series plots, ADF checks, and sequential OLS regressions. The final specification includes inflation lags at months 1, 2, and 12, one-month-lagged oil volatility, and contemporaneous USD/INR change.

Newey–West HAC standard errors are reported using a Bartlett kernel and bandwidth 4 because residual autocorrelation remains. HAC adjusts inference but does not repair model misspecification, omitted-variable bias, or establish causality.

4. Main Empirical Results

The baseline model has low explanatory power ($R^2 \approx 0.003$). Adding lagged inflation raises R^2 to approximately 0.81, indicating strong inflation persistence. Contemporaneous oil-price changes are not statistically significant in the tested specifications.

The contemporaneous oil-volatility coefficient is –0.00225 (p = 0.8003). In the final model, the one-month-lagged volatility coefficient is +0.015053; its conventional OLS p-value is 0.0909, while its Gretl HAC p-value is 0.0046. The latter supports a positive conditional association at the 5% level under the chosen HAC specification.

5. Final Model Results

Variable	Coefficient	HAC SE	HAC p-value	Interpretation
Constant	0.880365	0.308116	0.0050	Significant
Inflation lag 1	1.08629	0.086113	<0.001	Positive persistence
Inflation lag 2	–0.220535	0.085845	0.0114	Negative association
Inflation lag 12	–0.072007	0.037902	0.0597	Not significant at 5%
Oil volatility lag 1	0.015053	0.005223	0.0046	Positive; significant
USD/INR change	–0.017672	0.060050	0.7690	Not significant

Model fit: $R^2 = 0.8144$; adjusted $R^2 = 0.8071$; n = 132; AIC = 286.5968; BIC = 303.8936.

6. Diagnostics and Limitations

The White test has p = 0.3187 and does not reject homoskedasticity at conventional levels. The Breusch–Godfrey test rejects no residual autocorrelation up to order 12 (LM p = 0.0000728; F-test p = 0.000024); the Ljung–Box test is also strongly significant at lag 12.

The ADF result for CPI inflation is borderline and sensitive to lag specification. Potential omitted factors include food prices, monetary policy, domestic fuel-price adjustments, taxes, and broader supply shocks. HAC is a partial correction for standard errors; it does not remove serial dependence or establish causal direction.

7. Conclusion

The evidence is suggestive but not definitive. Contemporaneous oil-price changes and oil volatility are not statistically significant in the tested models. One-month-lagged oil volatility is positively associated with Indian CPI inflation under Newey–West HAC inference ($\beta = 0.0151$; $p = 0.0046$). Because residual autocorrelation persists and important controls are omitted, this should not be described as a causal effect or fully robust result.

References and Data Notes

1. Petroleum Planning & Analysis Cell (PPAC), International Prices of Crude Oil / Indian Basket: https://ppac.gov.in/index.php/prices/international-prices-of-crude-oil

2. Reserve Bank of India, DBIE, Consumer Price Index, Combined General Index (Base 2012 = 100): https://dev.dbie.rbihub.in/prices?table=real_sector.r45_consumer_price_index

3. Reserve Bank of India, DBIE, Average forex rate (monthly), unit Indian rupee, select UNITED STATES DOLLAR: https://dev.dbie.rbihub.in/statistics?table=financial_markets.avg_for_rat_rn

The exchange-rate series is the monthly-average USD/INR rate (rupees per US dollar), not the RBI's foreign-currency purchase/sale transactions. Confirm the currency selection and unit when downloading.

Newey, W. K., & West, K. D. (1987). A Simple, Positive Semi-definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. Econometrica, 55(3), 703–708.

Access date: 13 October 2026. Insert the exact source file names and download dates before final submission.